

# RAHUL MITRA

## ENGINEER - SOFTWARE, AI/ML, OPERATIONS RESEARCH & SECURITY

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### PROFILE SUMMARY

Early-career engineer and current NUS undergraduate (BE Industrial Systems Engineering, 2nd Major Computer Science, Minor Mathematics; graduating Jul 2027) working across the full technical stack: applied AI/ML (RAG, AI agents, fine-tuned transformers, HPC), solution architecture and cloud, operations research and optimization, and adversarial security. Proven through hands-on delivery: led the fine-tuning of a 109M-parameter BERT model on the ASPIRE 2A supercomputer, maintains the AsyncDDGS PyPI library, and shipped a Singpass/Myinfo-integrated civic balloting platform now production-ready for national rollout and winner of the S\$20,000 Sparks Community Innovation Fund. Also an active bug bounty researcher on YesWeHack against Singapore government and transport targets.

### EDUCATION

#### National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Ranked 1st of 24 students in 3D Computer Vision (AY25/26 Sem 2), earning the School of Computing Certificate of Outstanding Performance.
- Relevant coursework: Software Engineering, Artificial Intelligence, 3D Computer Vision, Data Structures and Algorithms, Simulation, Statistics, Operations Research, Linear Algebra, Probability, Computer Systems, Mathematics, Engineering Systems, Industrial Systems Engineering.

#### Anderson Serangoon Junior College (ASRJC)

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

### WORK AND INTERNSHIP EXPERIENCE

#### Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.
- Strengthen monitoring and process structure across healthcare supply chains with AI, helping ensure reliable product availability for vulnerable patient groups.

#### Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Jun 2026

- Built a Python digital-twin simulation and constraint-programming optimizer for an NP-hard hybrid flow-shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.

#### Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Awarded the S\$20,000 Sparks Community Innovation Fund for building an end-to-end digital garden-plot balloting platform (project codename Churp) spanning admin and public-facing frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC authorization-code flow / Relying Party) and Myinfo (v5) for passwordless authentication and consent-based retrieval of verified identity (NRIC/FIN) and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders.

#### Independent Bug Bounty Researcher, YesWeHack Platform

May 2024 - Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports using Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.
- Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.

#### Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

### SELECTED PROJECTS AND COMPETITIONS

#### Maritime Deficiency Severity Forecasting - Maritime Hackathon 2025 (Team Lead) - Hackathon

2025

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with additional DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer to forecast inspection deficiency severity.

#### Volt Pulse SG - SMU Hack For Cities 2026 (Top 8 finalist of 50+ teams) - Hackathon

Jan 2026

- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

#### AsyncDDGS - Open-Source Search Client - PyPI library

2024 - Present

- Authored and maintain an asyncio-first, aiohttp-based DuckDuckGo client published on PyPI with pytest CI/CD, adopted across AI tooling, Discord bots, and search projects.

#### AgeWellLah.AI - NUS HealthHack 2025 (AI/ML Engineer) - Hackathon

2025

- Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.

## SmartExam - GovTech x NTUPC Product Hackathon - Hackathon

2025

- Built a multi-agent autonomous exam-generation system with a RAG pipeline and Next.js frontend, automating source ingestion, question generation, and education workflows.

## EthosLens - LifeHack 2025 (Backend Developer) - Hackathon

2025

- Built an AI research engine scoring brand ethics and sustainability using FastAPI, Python, and LangChain with automated evidence retrieval.

## DSTA BRINHACK 2025 CTF (top quartile of 400+ teams) - Capture-the-flag competition (first CTF)

2025

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, finishing in the top quartile among 400+ teams.

## OnTheSpectrum - Codex & Blender MCP 3D Pipeline - Personal

May 2026

- Built a local-first Blender-to-Three.js asset pipeline using Codex and Blender MCP, generating GLB assets, metadata, previews, playable-world QA, and Remotion video workflows.

## IE2110 Operations Research Graph Optimization - Coursework

2024

- Built LP, network-flow, and nonlinear programming models, including a Jupyter SSSP/APSP (Dijkstra) graph optimization for decision support.

## Arcane Security Tooling Family - Personal

Mar 2026

- Built Rust-forward security and red-team tooling (Arcane, Arcane-PP, Arcane-GLM, Arcane-OCR) for automation, OCR, and adversarial workflows.

## Additional projects - Personal and coursework

2023 - 2026

- Project Utopia - real-time global situational-awareness dashboard with AI news aggregation, geopolitical monitoring, and geospatial visualization (Mar 2026).
- Interactive Portfolio Infrastructure - AI-assisted portfolio in React, TypeScript, Three.js, and Matter.js with profile switching, searchable field notes, and analytics (2026).
- NVIDIA Disaster Risk Monitoring with Satellite Imagery - geospatial AI with U-Net semantic segmentation on GPU/ASPIRE 2A workflows (2023).
- Fine-tuning LLMs for Cybersecurity - coursework exploring Mistral/LLaMA fine-tuning with AutoTrain and AutoGen (2024).

## TECHNICAL SKILLS AND CERTIFICATIONS

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**Languages:** Python, TypeScript, JavaScript, Java, Rust, Bash, C++, C#, Excel VBA, SQL

**AI/ML:** BERT, DNN, Transformers, RAG / RAG Architecture, AI Agents, OpenAI GPT-4, Gemini, SEALION, LangChain, Vector Search, Computer Vision, Model Evaluation, NLP, Deep Reinforcement Learning (PPO, A2C, DDPG, DQN, Double DQN), ASPIRE 2A / HPC / CUDA / distributed training

**Operations Research / Optimization:** Constraint Programming, Scheduling Optimization, Hybrid Flow Shop, Digital Twin Simulation, Simulation, Dijkstra / SSSP / APSP, Decision Analytics, Statistics, Jupyter, Mathematical Modeling (Linear Algebra, Probability, Calculus)

**Cloud / DevOps:** Azure, Docker, CI/CD, Bitbucket CI/CD, GitHub Actions, Supabase, API Design, Codex

**Security:** Burp Suite, Wireshark, SSRF, CSRF, SQL Injection, NoSQL Injection, Authentication Bypass, OAuth2/JWT, OWASP Top 10, Secure Design, Fuzzing, Reverse Engineering, Custom Exploit Scripting, Cloud Security (AWS IMDSv2, Metadata APIs)

**Web / Full-Stack:** React, Next.js, Three.js, Matter.js, FastAPI, Flask, aiohttp, asyncio, GIS Mapping, Singpass Login & Myinfo (OIDC), Konva.js

**Solution Architecture / Product:** Solution Architecture, Manufacturing Systems, Stakeholder Discovery, Rapid Prototyping, Accessibility, Product Discovery, User Testing, Dashboards

**Hardware / Embedded:** Edge AI / Hailo model deployment, Autonomous robotics (sensor integration)

- Disaster Risk Monitoring Using Satellite Imagery - NVIDIA Deep Learning Institute (Dec 2023)
- Deep Reinforcement Learning Hands-On (DQN, Double DQN, DDPG, PPO, A2C) - Packt (2023)
- Docker for the Absolute Beginner - Hands-On - Stone River eLearning (2024)
- Reverse Engineering Windows Executables (digital forensics) (2024)
- Excel 2019 Advanced - Intellezy (2023)
- Deep Learning CNN with Python - Packt (2023)
- Deep Learning RNN with Python - Packt (2023)
- Learning C# by Developing Games with Unity - Packt (2023)

## CO-CURRICULAR AND COMMUNITY

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### Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Support senior engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

### Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

Mar 2019 - Aug 2020

- Led astronomy outreach and organized weekly stargazing and observation programs.
- Competed in the 7th Singapore Astronomical Olympiad and engineered an autonomous robot for the National Robotics Challenge.
- Received the ASRJC Outstanding Contribution Award.